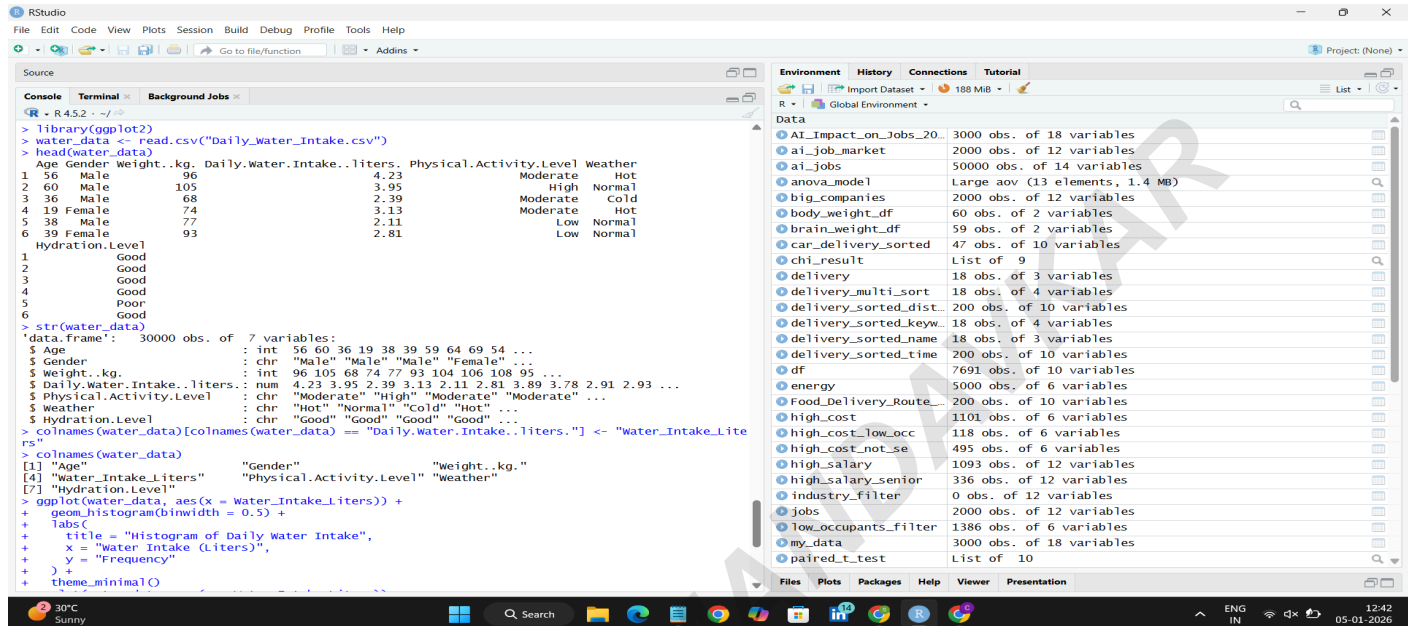


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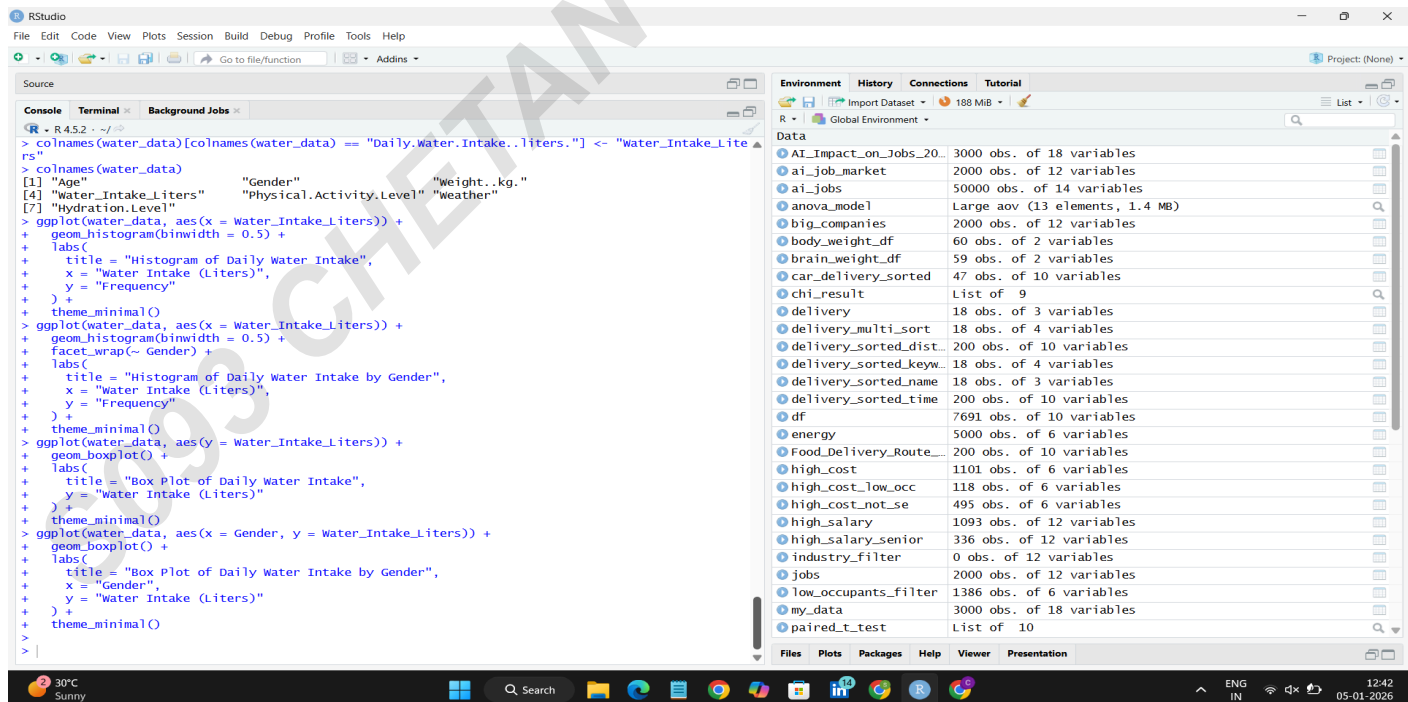
Aim:- Generating histograms and box plots using ggplot2 (R).



The screenshot shows the RStudio interface. The console on the left contains the following R code:

```
R - R4.5.2 - ~/ -  
> library(ggplot2)  
> water_data <- read.csv("Daily_Water_Intake.csv")  
> head(water_data)  
  Age Gender Weight..kg. Daily.Water.Intake..liters. Physical.Activity.Level Weather  
1  56   Male      96           4.23           Moderate      Hot  
2  60   Male     105           3.95           High      Normal  
3  36   Male      68           2.39           Moderate      Cold  
4  19  Female      74           3.13           Moderate      Hot  
5  38   Male      77           2.11           Low      Normal  
6  39  Female      93           2.81           Low      Normal  
  Hydration.Level  
1      Good  
2      Good  
3      Good  
4      Good  
5      Poor  
6      Good  
> str(water_data)  
*data.frame*: 30000 obs. of  7 variables:  
 $ Age          : int  56 60 36 19 38 39 59 64 69 54 ...  
 $ Gender       : chr  "Male" "Male" "Male" "Female" ...  
 $ Weight..kg.  : int  96 105 68 74 77 93 104 106 108 95 ...  
 $ Daily.Water.Intake..liters.: num  4.23 3.95 2.39 3.13 2.11 2.81 3.89 3.78 2.91 2.93 ...  
 $ Physical.Activity.Level : chr  "Moderate" "High" "Moderate" "Moderate" ...  
 $ Weather      : chr  "Hot" "Normal" "Cold" "Hot" ...  
 $ Hydration.Level : chr  "Good" "Good" "Good" "Good" ...  
> colnames(water_data)[colnames(water_data) == "Daily.Water.Intake..liters."] <- "Water_Intake_Liters"  
> colnames(water_data)  
[1] "Age" "Gender" "Weight..kg." "Water_Intake_Liters" "Physical.Activity.Level" "Weather" "Hydration.Level"  
> ggplot(water_data, aes(x = Water_Intake_Liters)) +  
+   geom_histogram(binwidth = 0.5) +  
+   labs(  
+     title = "Histogram of Daily Water Intake",  
+     x = "Water Intake (Liters)",  
+     y = "Frequency"  
+   ) +  
+   theme_minimal()
```

The Environment pane on the right shows the loaded data frame 'water_data' with 30000 observations and 7 variables.



The screenshot shows the RStudio interface. The console on the left contains the following R code:

```
R - R4.5.2 - ~/ -  
> colnames(water_data)[colnames(water_data) == "Daily.Water.Intake..liters."] <- "Water_Intake_Liters"  
> colnames(water_data)  
[1] "Age" "Gender" "Weight..kg." "Water_Intake_Liters" "Physical.Activity.Level" "Weather" "Hydration.Level"  
> ggplot(water_data, aes(x = Water_Intake_Liters)) +  
+   geom_histogram(binwidth = 0.5) +  
+   labs(  
+     title = "Histogram of Daily Water Intake",  
+     x = "Water Intake (Liters)",  
+     y = "Frequency"  
+   ) +  
+   theme_minimal()  
> ggplot(water_data, aes(x = Water_Intake_Liters)) +  
+   geom_histogram(binwidth = 0.5) +  
+   facet_wrap(~ Gender) +  
+   labs(  
+     title = "Histogram of Daily Water Intake by Gender",  
+     x = "Water Intake (Liters)",  
+     y = "Frequency"  
+   ) +  
+   theme_minimal()  
> ggplot(water_data, aes(y = Water_Intake_Liters)) +  
+   geom_boxplot() +  
+   labs(  
+     title = "Box Plot of Daily Water Intake",  
+     y = "Water Intake (Liters)"  
+   ) +  
+   theme_minimal()  
> ggplot(water_data, aes(x = Gender, y = Water_Intake_Liters)) +  
+   geom_boxplot() +  
+   labs(  
+     title = "Box Plot of Daily Water Intake by Gender",  
+     x = "Gender",  
+     y = "Water Intake (Liters)"  
+   ) +  
+   theme_minimal()
```

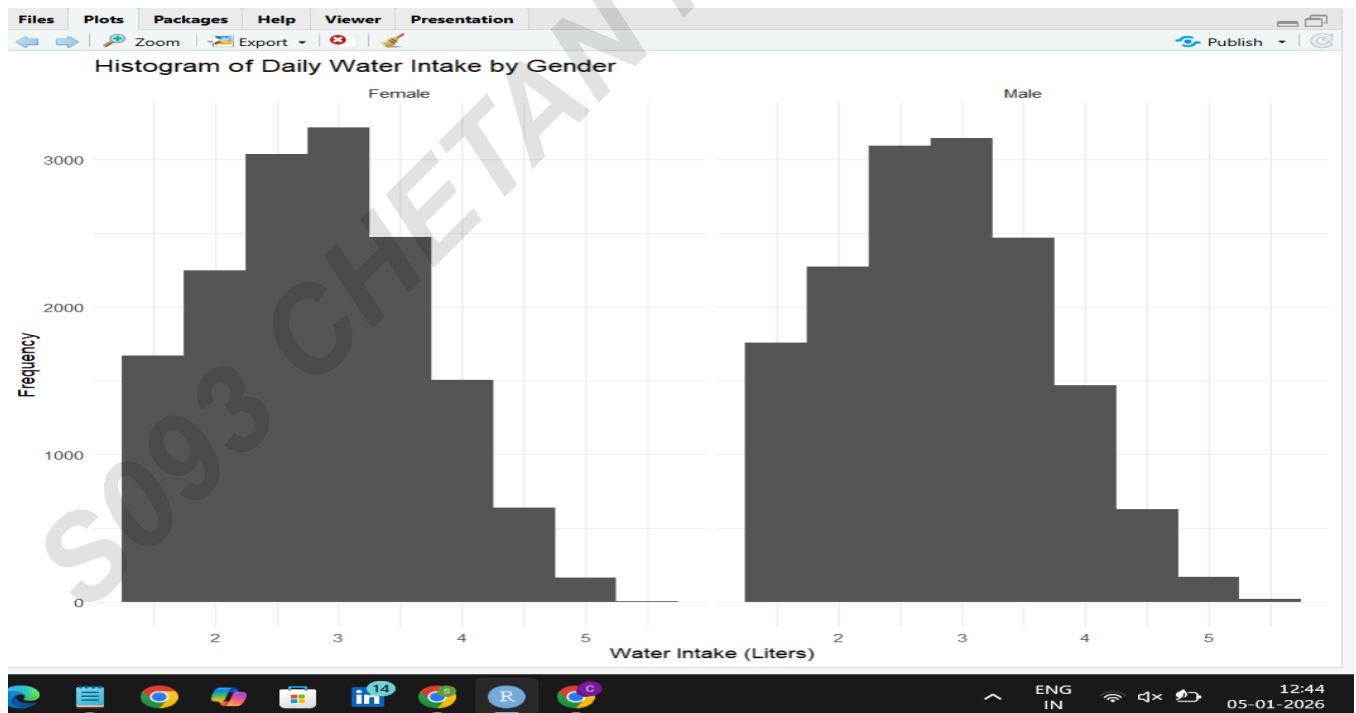
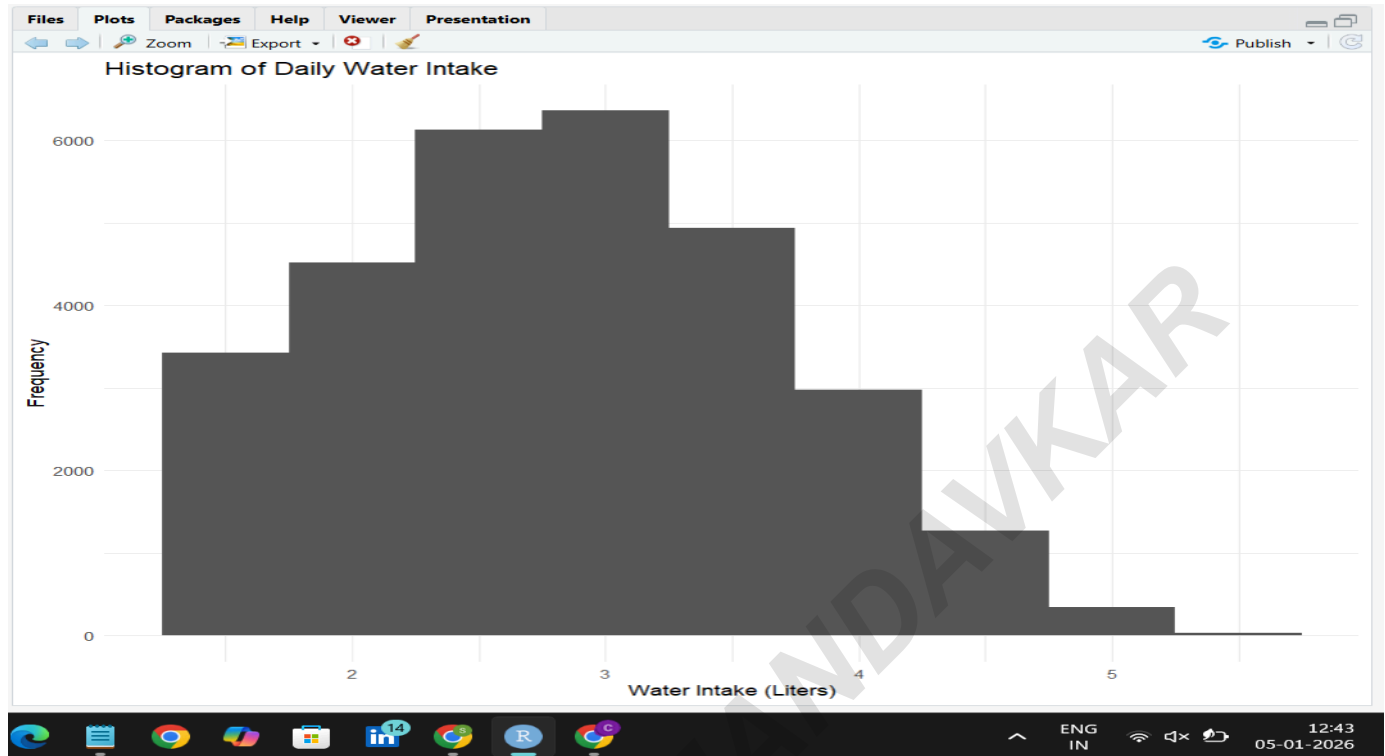
The Environment pane on the right shows the loaded data frame 'water_data' with 30000 observations and 7 variables.

NAME:- CHETAN MANDAVKAR

ROLL NO. S093

SUBJECT:- Data Analysis With SAS / SPSS / R

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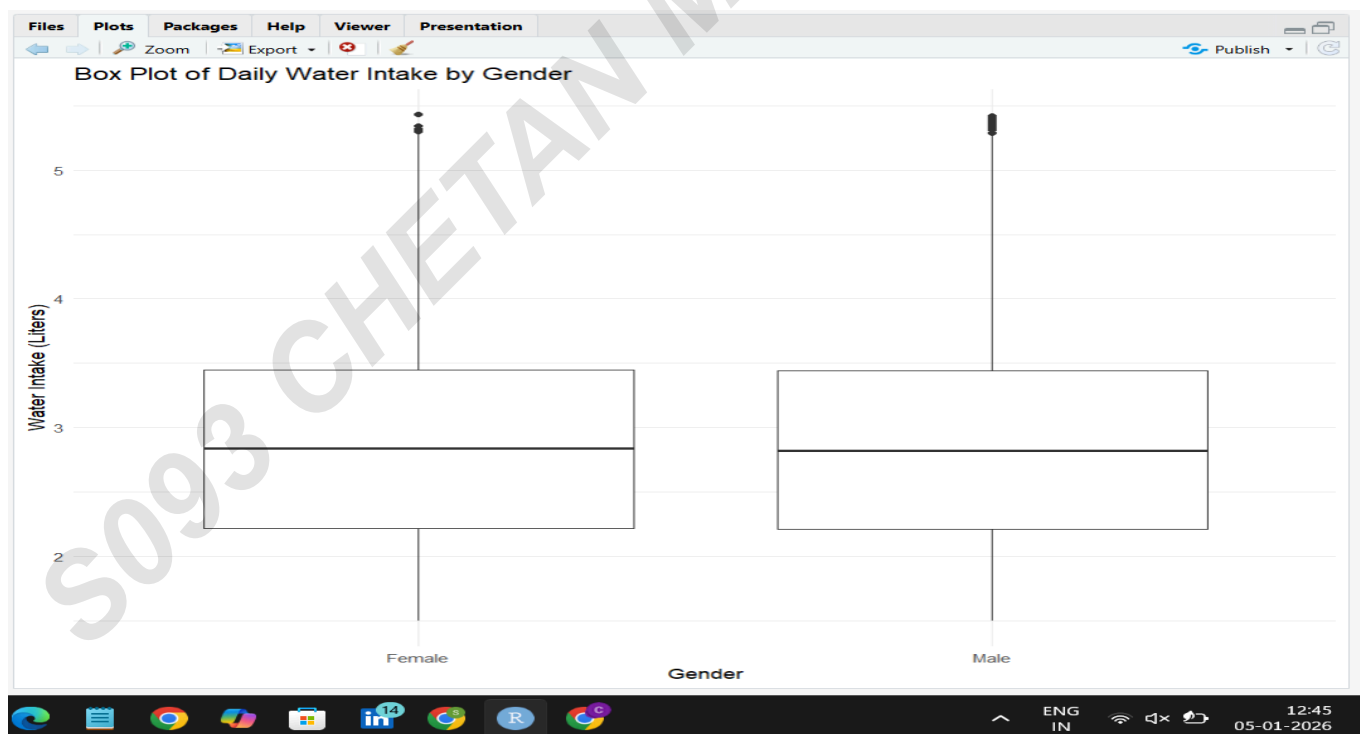
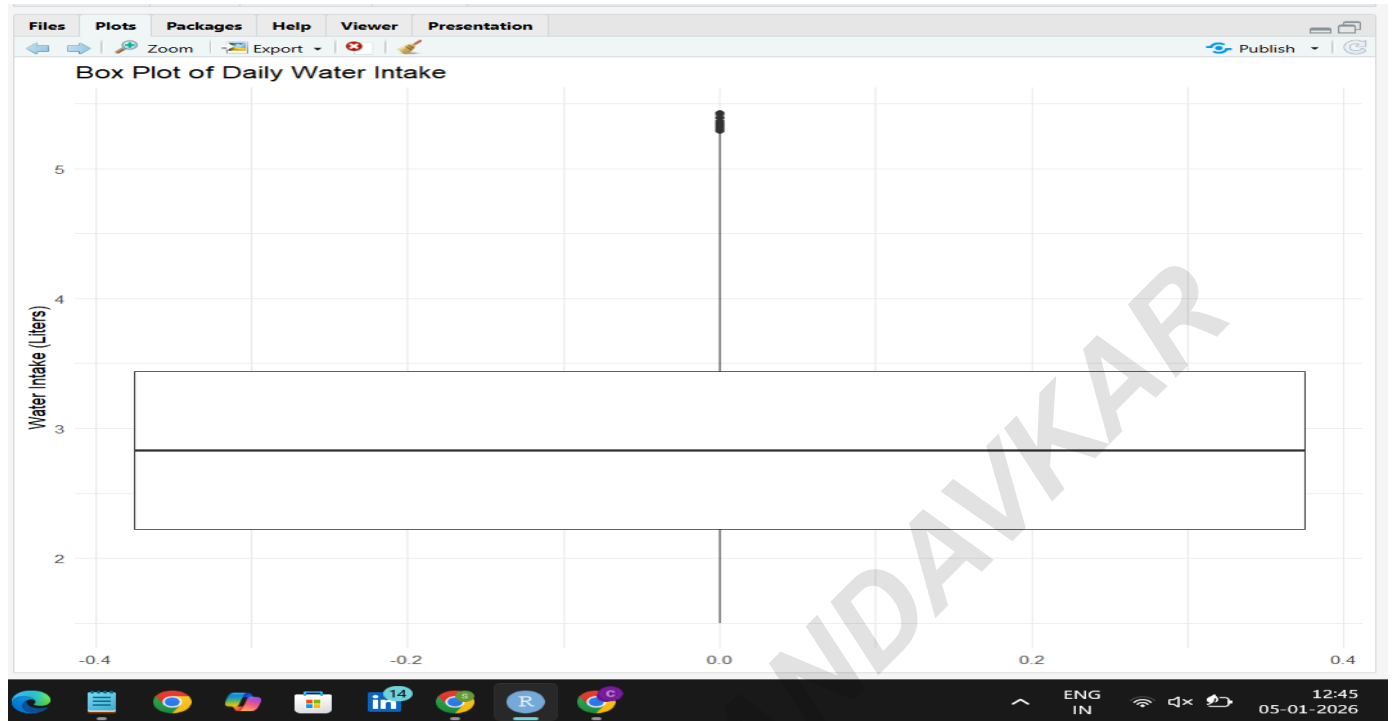


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